

EVSP DR Current Research

Week of 14 September. Results checked 12 September 2026 at 03:25 EDT; overnight extensions are still running. Open [Figures with explanations](#) for timing, charging costs and geography, or [CG curves and bus schedules](#) for the restored figure library.

Current finding. With set covering and inherited columns, all six chains match the GIRO fleet at $k=3-6$. Chains 3, 5 and 6 also reach ten buses at $k=10$. The new 512-route import treatment reaches 11 buses at $k=11$ in chain 6; other completed extension cases need extra buses.

Integer fleet with full-pool inheritance

Each cell is the number of integer bus routes found. Green matches the target. Red requires extra buses. A dash means no completed result.

Target buses	Chain 1	Chain 2	Chain 3	Chain 4	Chain 5	Chain 6
2	2	2	2	2	3	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	—	7	7	7	7	7
8	—	8	8	8	8	8
9	—	—	9	9	9	9
10	—	—	10	—	10	10

The table above retains the earlier full-pool treatment. The extension checks at most 512 inherited routes before CG; its new integer results are shown separately below. Runs continue through $k=15$ for all six chains.

$k=2$ starts from single-trip routes; later sizes reuse validated previous- k columns. Chain 3 $k=10$ also includes routes from a fresh solver solution. These are pool proofs. Chain 2 $k=10$ (71 buses) had zero pricing iterations and no CG certificate. Individual route replay passed; duplicate-trip removal and shared charger capacity are not established by this table.

Fresh starts on the same chains

Each size starts independently. Both tables use covering and 240-kWh batteries with 240-kW charging. Fresh fleet proof status varies by cell.

Target buses	Chain 1	Chain 2	Chain 3	Chain 4	Chain 5	Chain 6
5	5	5	5	5	6	5
8	9	9	9	9	9	8
10	11	11	11	11	11	11
15	18	17	18	19	16	19

Inherited columns improve the available integer pool: chain 3 at $k=8$ improves from a proved nine-bus pool optimum to eight; chain 5 at $k=5$ improves from a proved six-bus pool optimum to five.

What is limiting progress

Checking saved routes took too long. Before starting CG on a larger trip set, we check that routes saved from the previous size are still valid. In chain 1 at $k=7$, chain 2 at $k=9$ and chain 4 at $k=10$, this checking used up the available time. We now check at most 512 saved routes, for at most 15 minutes, then let CG search for new routes. All three cases have now finished CG. Their integer solves found 8, 10 and 11 buses, respectively; only the first and third are proved minimums within their saved pools. Using fewer saved routes may affect the final integer solution, so this is a separate experiment.

Charger limits made the search for new routes very slow. In one three-bus test, the LP solved in 0.006 seconds, but one search for a new route took 7.15 hours. The run ran out of time before it could prove that no improving route remained. Its 16-bus integer solution uses only the routes found so far; it does not show that 16 buses are necessary. The timing figures explain these two different delays.

Case	Baseline	PARX 60 kW	Station capacity	Both
$k=1$, two duties tested	1 each	1 each	1 each	1 each
$k=2$, 23 trips	2	2	3	3
$k=3$, 35 trips	3	3	16	16

Sixteen is the result in a small generated pool, not a proved physical requirement.

Capacity-constrained schedules passed station sweeps. This pilot has no 65% return-SOC floor and is separate from the six baseline chains.

Overnight extension results

At 03:25 EDT, **45 of 87 CG cases are certified** (17 warm, 28 decomposition groups) and 19 MIPs have finished. Runs continue through k=15 and across ten decompositions of a 32-duty parent.

Warm results with at most 512 inherited routes. All 12 CG results below are certified at reduced-cost tolerance 0.0001. CG minutes include initialization. Fractional route weight equals target; the weighted objective also includes charging costs.

Case k = target	Buses	Pool fleet bound	Fleet proved in pool	CG minutes	Weighted LP objective
Chain 1, k=7	8	8	Yes	53.6	700313.457
Chain 1, k=8	9	9	Yes	49.5	800383.688
Chain 2, k=9	10	9	No	64.0	900301.353
Chain 2, k=10	11	10	No	55.7	1000371.146
Chain 3, k=11	12	11	No	27.7	1100375.859
Chain 3, k=12	14	12	No	32.8	1200427.901
Chain 3, k=13	14	13	No	52.6	1300474.559
Chain 4, k=10	11	11	Yes	42.0	1000426.271
Chain 4, k=11	12	11	No	59.4	1100462.865
Chain 5, k=11	12	11	No	70.5	1100522.228
Chain 6, k=11	11	11	Yes	32.5	1100399.029
Chain 6, k=12	13	12	No	64.6	1200464.548

The fleet bound concerns saved columns only. Every charging-stage solve reached its time limit. For chain 1 k7/k8 and chain 4 k10, the fleet minimum is proved above target; different columns are needed to reach the target. Other above-target gaps remain unresolved.

Decomposition: four of seven completed groups match 8 buses; three use 9. No combined 32-duty result yet. All 19 solutions pass individual-route replay and coverage; duplicate removal and shared charger capacity remain unchecked.

Default MIPs: the earlier 75 and new 19 completed with zero confirmed preemptions. Eleven new MIPs are running; their outcomes remain unknown.

Methods and evidence

CG route cost = 100,000 + electricity + 5 per charging start. MIP stage 1 minimizes buses. Stage 2 constrains buses $\leq$ the validated stage-1 incumbent and minimizes electricity plus start fees.

A pricing certificate establishes the LP result only in its represented graph and tolerance. A MIP proof concerns its saved columns. Fleet attainment, physical replay and shared-capacity validation are separate checks.

[Experiment register](#) · [Dated results and source tables](#) · [Experiment settings](#) · [Historical document and figures](#) · [Local prototype tests and charging-cost bug](#)

Storage, 12 September: 156 cold files archived losslessly; 133.74 GB freed. Active research and V2G data preserved. [Restore instructions](#).

Paired speed tests: the five original fresh/capacity jobs and four warm replacements remain active. One reference arm has finished CG; no complete pair or speedup result yet. The original four launcher failures remain recorded. [Jobs, settings and recovery evidence](#).